



河北山启新材料科技有限公司

Hebei Thanky New Materials Techonlogy Co.,Ltd.



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Graphite Products



Properties	Unit	Standards/CNC			Standards/Molding
Bulk Density	g/cm ³	1.63	1.7	1.8	1.8±0.1
Resistivity	μΩm	11	9	8.5	12
Ash	PPM	3000	800	500	700
Coefficient of Thermal Expansion	10 ⁻⁶ /°C	—	2.5	4.75	4.55
Compression Strength	Mpa	>20	30	65	65
Flexural Strength	Mpa	>9	15	35	35
Thermal Conductivity	W/m.k	—	125	121	—
Apparent Porosity	%	—	15	13	—
Hardness	HSD	—	30	40	—
Partial Size	mm	2-4	0.8-2	0.014	0.05

- **High-Temperature Stability:** Performs reliably at temperature above 2000°C.
- **High Thermal Conductivity:** Ensures uniform heating of materials and improves production efficiency.
- **High thermal Shock Resistance:** Minimizes the risk of deformation and cracking in extreme thermal conditions.

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Mullite Products



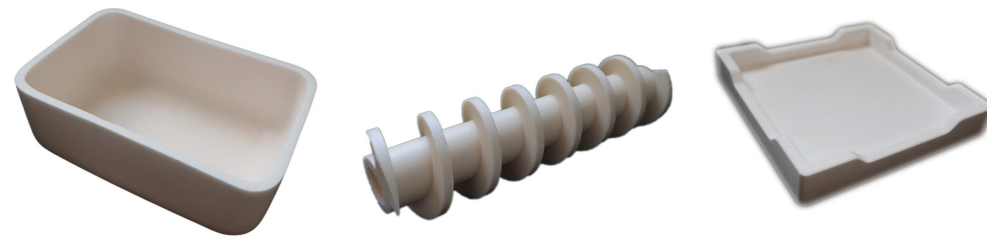
Product Categories		Unit	Saggar+Coating			Corundum
			Cordierite Mullite	Corundum Cordierite	Coating	
Chemical Analysis	Al ₂ O ₃	%	34-37	60	25.74	>93
	SiO ₂	%	52-53	24	56.26	—
	MgO	%	7.5-8	15	10.02	—
	Fe ₂ O ₃	%	0.4-1	0.2	—	—
	ZrO ₂	%	—	—	5.58	—
Physical properties	Density	g/cm ³	1.85-1.9	2.1	—	2.6
	Coefficient of Thermal Expansion	×10 ⁻⁶ /K (20-1000°C)	2.2-3.0	4.2	—	4.5
	Max Operating Temperature	°C	1280	1320	—	1550
	Flexural Strength at 20°C	MPa	10-15	8	—	18

- **High-temperature resistance:**Used in temperature below 1600°C.
- **High thermal shock resistance:**Resistant to cracking or peeling off under rapid temperature changing.
- **High Chemical Stability:** Maintain stability in chemical environments,resistant to chemical reactions in corrosive conditions.



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Alumina Products

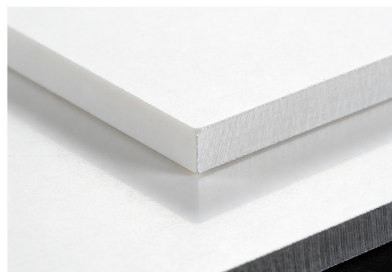
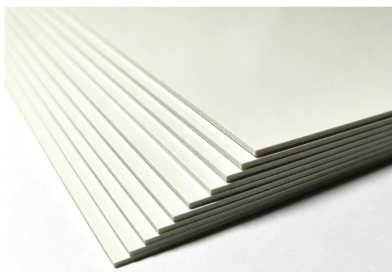


Items	Unit	Index
Al ₂ O ₃	%	99.7
Bulk Density	g/cm ³	3.92
Flexural Strength	MPa	380
Coefficient of Linear Expansion	×10 ⁻⁶ /°C (25-800°C)	7.6
Thermal Conductivity	W/m·k	35
Dielectric Strength	KV/mm	10
Long-term Use Temperature	°C	1600

- Long life span
- Good thermal stability
- Good peeling resistance
- High compressive strength
- Flat surface,Uniform color
- Excellent bottom flatness
- Strong corrosion resistance
- No cracks,falling edges,corners or pits in appearance



Ceramic Substrates



Items		Unit	Al ₂ O ₃			
Content		%	96	99	99.6	99.9
Color		-	White	Off-white	White	White
Surface Roughness		μm	0.2-0.6	0.15-0.4	0.1-0.3	0.1-0.2
Density		g/cm ³	≥3.78	≥3.8	≥3.85	≥3.9
Reflectivity		%	92	85	83	/
Mechanical Properties	Flexural Strength	MPa	>380	>400	>470	>500
	Fracture Toughness	MPa·m ^{1/2}	3	3.2	3.2	3.2
	Vickers Hardness	GPa	14	15	16	16
	Young's Modulus	GPa	330	350	350	350
Thermal Properties	Coefficient of Thermal Expansion	×10 ⁻⁶ /k	7.8	7.9	7.9	7.9
	Thermal Conductivity	W/m·k	>24	>28	>29	>29
	Specific Heat Capacity	J(kg/k)	750	780	780	780
Electrical Properties	Dielectric Constant	-	9.4	9.8	9.8	9.8
	Dissipation Factor	×10 ⁻⁴	≤3	≤3	≤2	≤2
	Volume Resistivity	Ω·cm	≥10 ¹⁴	≥10 ¹⁵	≥10 ¹⁵	≥10 ¹⁶
	Dielectric Strength	KV/mm	≥15	≥15	≥15	≥15

- **High Strength:** Hard and stable, providing mechanical support and protection for delicate chips.
- **High Thermal Conductivity:** Quickly dissipates heat to ensure stable operation of components at high power.
- **Insulating Safety:** Naturally excellent insulator, effectively isolates electrical potential and prevents short circuits.



Alumina Ceramic Tubes



Items	Unit	Thanky	Competitor	Competitor	Thanky	Competitor	Competitor
Al ₂ O ₃	%	99	99	99.5	99.7	99.7	99.9
Bulk Density	g/cm ³	3.81	3.8	3.9	3.85	3.9	3.9
Flexural Strength	MPa	340	310	370	370	380	400
Coefficient of Linear Expansion	×10 ⁻⁶ /°C (25-800°C)	7.6	/	/	7.6	/	/
Thermal Conductivity	W/m·k	34	29	32	35	32	34
Dielectric Strength	KV/mm	10	15	15	10	15	15
Long-term Use Temperature	°C	1600	/	/	1700	/	/

- **Smooth Internal Surface:** Minimizes fluid resistance, ideal for transporting high-purity media.
- **Hardness & Wear Resistance:** Excellent hardness and wear resistance for high abrasion environments.
- **Mechanical Strength:** High flexural strength, excellent compressive performance with extended service life.



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'TORAY'

Innovation by Chemistry

Current series of Beads



Rare-earth-free Beads



Composition	Unit	Content
ZrO ₂	%	94.75±0.60
Y ₂ O ₃	%	4.90±0.50
Al ₂ O ₃	%	0.30±0.10
Fe ₂ O ₃	%	≤0.01
SiO ₂	%	≤0.01
TiO ₂	%	≤0.01
Items	Unit	Index
Density	g/cm ³	6.06
Bulk Density	g/cm ³	3.7
Hardness	Hv	1300

	Current series	Rare-earth-free series
Main component	ZrO ₂ standard	ZrO ₂ Low-temperature sintering
Stabilizer	Y ₂ O ₃ Rare-earth element	Non-rare earth element
sintering temperature	1500°C	1250°C

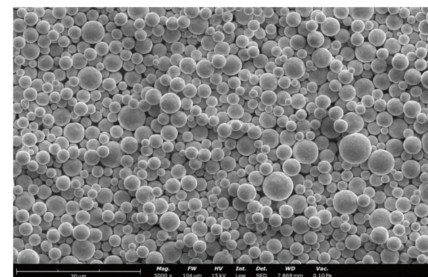
- High strength, high density, high hardness.
- Excellent wear and corrosion resistance.
- Uniform distribution of microstructure, high dispersion performance.



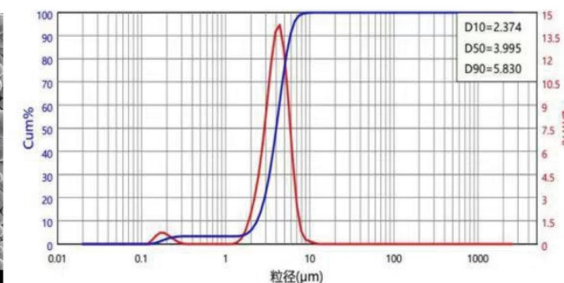
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Alloy Nickel Powder

SEM



Particle Size Distribution Curve



(Take Ni4000 alloy nickel powder as an example)

Items		Unit	Result		
			Ni3000	Ni4000	Ni5000
PSD	D10	μ m	1.791	2.374	2.179
	D50	μ m	3.110	3.995	4.756
	D90	μ m	4.858	5.830	8.495
	D99	μ m	6.456	7.777	13.200
TAP		g/cm ³	5.3	5.0	5.1
H ₂ O		PPM	117	500	600
Chemical Composition	O	PPM	1200	629	425
	C	PPM	102	125	92
	S	PPM	15	18	30

- High uniformity, high stability.
- Narrow particle size distribution, few large particles.
- High sphericity, no satellite particles, no irregular shapes.
- Low level of impurities and foreign particles, with controlled oxygen and carbon content.